

LC877 : Stone Game

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Adobe

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Problem Setup & Rules



n piles, where *n* is even

- Alice starts first, can take stones from end or first.
- Person with most stone wins.
- If Aadya won → **true**, else → **false**.

3D DP Approach

State Representation:

$dp[l][r][0]$ or $dp[l][r][1]$

- The $dp[l][r][0]$ stores the max possible score the current player can get.
- $dp[l][r][1]$ max poss score other player can get.

Then for every len, l, r, we make DP table.

Lets say currplayer took left pile then after taking it, currplayer becomes otherplayer, and other becomes currplayer.

Taking Left Pile:

```
int currleft = piles[l] + dp[l+1][r][1]
```

```
int otherleft = dp[l+1][r][0]
```

Similarly for right

```
if (currleft > currright) {  
    dp[l][r][0] = currleft;  
    dp[l][r][1] = otherleft;  
}
```

C++ Implementation

```
class Solution {  
public:  
    bool stoneGame(vector<int>& piles) {  
        int n = piles.size();  
        vector<vector<vector<int>>> dp(n, vector<vector<int>>(n,  
vector<int>(2, 0)));  
  
        // dp[l][r][0] or dp[l][r][1]  
        // 0->current player  
        // 1->other player  
  
        for(int i = 0; i < n; i++){  
            dp[i][i][0] = piles[i];  
            dp[i][i][1] = 0;    // 1 hi pile he , 1 hi turn he ,  
current player ko milega sirf  
        }  
  
        for(int len = 2; len <= n; len++){  
            for(int l = 0; l + len - 1 < n; l++){  
                int r = l + len - 1;  
  
                // left pile utha liya curr ne  
                // uthane ke baad curr opponent ban gya
```

```

        // opponent curr ban gya
        int currlleft = piles[l] + dp[l+1][r][1];

        // curr ke left uthane ke baad , opponent ka optimal
        int otherleft = dp[l+1][r][0];

        int currrright = piles[r] + dp[l][r-1][1];
        int otherrright = dp[l][r-1][0];

        if(currlleft > currrright){
            dp[l][r][0] = currlleft;
            dp[l][r][1] = otherleft;
        }
        else{
            dp[l][r][0] = currrright;
            dp[l][r][1] = otherrright;
        }
    }
}

if(dp[0][n-1][0] > dp[0][n-1][1]) return true;
else return false;
}
};

```